

Physics-informed reconstruction of pulsatile flow

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An exploratory learning project connected to Control Principles (MECH3610)

Research question

Can a known dynamical model help reconstruct a flow field and estimate an unknown periodic input from sparse, noisy velocity measurements? This project uses an idealised fluid system to explore a question relevant to control: how much can we infer about a system when its measurements are incomplete?

System and unknown quantities

The system is a rigid circular tube with fully developed, axisymmetric, periodic flow. The dimensionless radius and cycle time lie in $[0, 1]$. The Womersley parameter is fixed at $\alpha = 3$. The velocity $u(r, t)$ obeys:

$$\frac{\alpha^2}{2\pi} \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + G(t), \quad G(t) = a_0 + a_c \cos(2\pi t) + a_s \sin(2\pi t)$$

The PINN jointly estimates the velocity field and the three forcing coefficients. Geometry, frequency and the harmonic form of G are known. The simulated reference uses $(a_0, a_c, a_s) = (6, 3.2, 2.4)$. These values generate the observations and evaluation reference, but do not enter the training targets for the equation.

Implementation and progression

The project began with the analytical Poiseuille profile $u(r) = 1 - r^2$, then introduced a neural approximation, a physics loss and inverse estimation. The final study adds time dependence and an interval with no velocity observations.

The Python implementation uses NumPy and PyTorch. A two-hidden-layer network with 32 units per layer takes r^2 , $\sin(2\pi t)$ and $\cos(2\pi t)$ as inputs. Multiplication by $(1 - r^2)$ imposes the wall condition, while the radial and temporal features impose centre symmetry and periodicity. Automatic differentiation supplies the PDE residual and wall derivative.

Scope of the work

This is offline field reconstruction and unknown-input estimation. It does not yet implement a state observer, feedback controller or closed-loop stability analysis. All measurements are simulated, with no patient data or medical validation.

Development disclosure: this is an AI-assisted learning project. AI supported code development and analysis. The results in this note come from the recorded experiments; the discussion distinguishes completed work from proposed extensions.

A paired reliability study

48 measurement scenarios: 4 sample counts x 4 noise levels x 3 seeds

The study varies the number of observations (12, 24, 48, 96) and Gaussian noise standard deviation (0%, 3%, 10%, 20% of the fixed velocity scale 1.5). Six observed phases remain fixed; no velocity measurement lies in 30-65% of the cycle. Increasing the count adds nested radial locations. Seeds 42-44 vary both noise and neural initialisation.

Three methods receive identical data: an inverse PINN, a neural network without the PDE loss, and analytical least squares using the correct harmonic solution family. Both networks share architecture, initial weights and boundary/periodicity constraints. Each uses 1,000 Adam steps and at most 100 L-BFGS iterations. Equal iteration limits do not imply equal convergence or runtime.

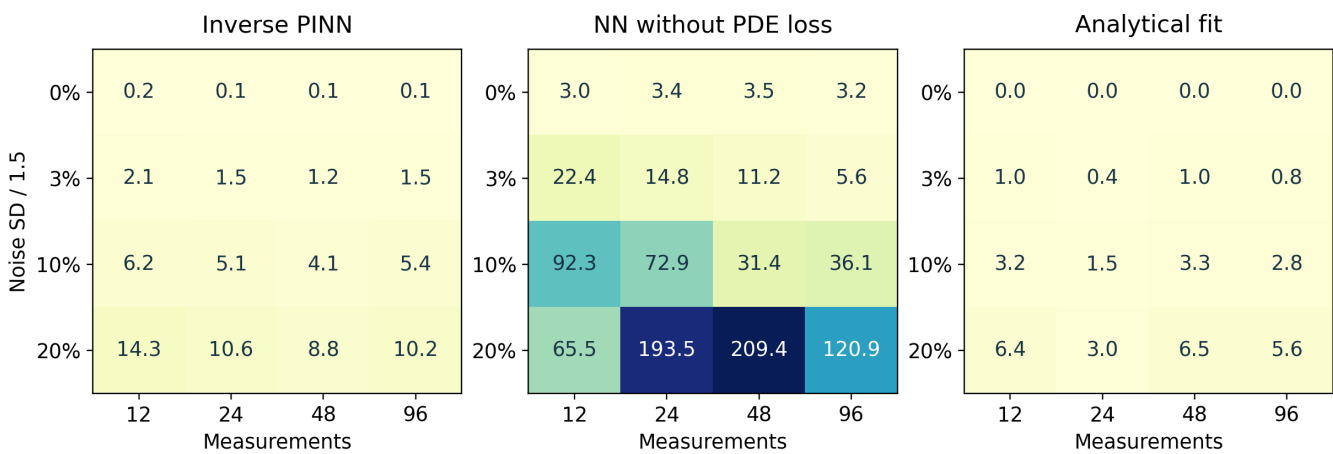


Figure 1. Mean hidden-interval velocity error (%) across three seeds. Error = RMSE / 1.5 x 100. Colours share one linear scale (0 to 209.4%) across all panels; darker means larger error. The displayed values are rounded.

Results and interpretation

Across all 16 configurations, the PINN has lower mean error than the neural baseline for each of the three metrics: hidden-interval velocity, full-cycle flow and full-cycle wall shear. Analytical least squares has lower mean error than the PINN in all 16 configurations for each metric.

48 observations, 10% noise	Velocity (%)	Flow (%)	Wall shear (%)
Inverse PINN	4.11 ± 4.55	4.27 ± 3.66	4.78 ± 4.63
NN without PDE loss	31.42 ± 2.49	22.94 ± 2.22	28.14 ± 6.16
Analytical fit	3.25 ± 2.39	3.56 ± 1.44	3.86 ± 1.60

Table: mean ± sample standard deviation over three seeds. Flow and signed wall-shear errors use full-cycle relative L2 norms. Evaluation uses 101 phases x 121 radii. Flow integrates the velocity with the radial weight; wall shear uses the negative wall derivative.

Main lesson: the PDE constraint improves this neural baseline, but the simpler correct-family fit remains preferable in this favourable synthetic setting. A physics-informed model should earn its complexity through comparison with classical estimation methods.

What I would like to investigate next

A link between the completed experiment and further study in MECH3610

MECH3610 covers system equations, state-space systems and system response [1]. This project gives me a concrete setting in which to explore those ideas: the velocity field evolves under a forcing input, while measurements reveal only part of that response. My interest is in understanding which estimation tools are appropriate before attempting feedback control.

A classical state-space comparison

A next step would be to discretise the radial diffusion operator and formulate a finite-dimensional state-space model. I would then compare its estimation of the unobserved velocity with the PINN on the same measurement sets. An observer or Kalman-based formulation would require explicit assumptions about process noise, measurement noise and treatment of the unknown forcing. This comparison has not yet been implemented.

Measurement placement and identifiability

The present analytical fit estimates three coefficients from a linear design matrix. Its rank and conditioning provide a tractable way to study when the chosen measurements carry enough information. I would like to connect this calculation with observability in a state-space model, and compare radial sampling with the addition of new observed phases.

Limitations that shape the next experiment

The benchmark assumes the correct geometry, frequency and forcing family. Three seeds give only an initial view of variability. A fixed training budget can leave optimisation error, so poor performance does not establish a fundamental limitation of a method. In particular, more measurements do not improve every noisy realisation. Recorded fitting times include concurrent CPU work and are not a controlled speed comparison.

A useful next test would deliberately perturb an assumed model parameter, such as α , while preserving the same observations. This would probe model mismatch rather than repeat the favourable correct-model comparison. No such result is claimed here.

Question for discussion

For this type of system, would a reduced state-space model with an observer be the most useful next step, or should measurement placement and identifiability come first? I would welcome guidance on which direction would best strengthen my understanding of control and estimation.

Reproducibility and references

The project preserves its source code, protocol and source hashes, all 48 sets of observations, model weights, predictions and metrics. It also includes an offline English explorer and a full numerical report. These materials are available for a follow-up discussion.

[1] HKUST, MECH3610: Control Principles, Fall 2026-27. [Official course description](#) (accessed 14 September 2026).

[2] M. Raissi, P. Perdikaris and G. E. Karniadakis (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378, 686-707. [doi:10.1016/j.jcp.2018.10.045](https://doi.org/10.1016/j.jcp.2018.10.045). Methodological background; the numerical results in this note come from the local project.